Boyao **Liao** | Curriculum Vitae

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RESEARCH INTERESTS

Learning theory, deep learning theory, optimization theory, stochastic gradient descent, Hessian spectral analysis, online learning, combinatorial multi-armed bandits, bandits with knapsacks, and online shortest-path routing.

EDUCATION

Jinan University – University of Birmingham Joint Institute **Guangzhou, China**
B.Sc. in Mathematics with Applied Mathematics *September 2023–Now*

- GPA: 4.11/4.25, rank: 3/68. *University of Birmingham criterion: above 75 is A+.*
- GPA: 4.18/5, rank: 3/68. *Jinan University criterion: 90–100 is A.*

AWARDS & HONORS

- **Best Student Paper Award**, The 37th International Conference on Algorithmic Learning Theory (ALT), 2026
- **National Scholarship**, Ministry of Education of the People's Republic of China, 2024
- **Outstanding Student**, Jinan University, 2024

RESEARCH EXPERIENCE

Optimization Dynamics and Learning Theory for Deep Neural Networks *2025–Now*

- Study fine-grained optimization dynamics of stochastic gradient descent, with an emphasis on step-size conditions, loss landscapes, and architecture-dependent training behavior.
- Analyze Hessian spectral phenomena in deep neural networks, including how depth influences bifurcation-like spectral behavior during training.
- Work under the supervision of Prof. Yaoqing Yang at Dartmouth College, with ongoing guidance from Prof. Hongyi Jiang at City University of Hong Kong.

Online Routing and Budgeted Multi-Armed Bandits *2025–Now*

- Study online shortest-path routing under hard anytime budget constraints in a combinatorial semi-bandit setting.
- Develop and analyze the Active Under-utilization and Skip for Anytime Knapsacks (AUSAK) algorithm, which combines lower-confidence-bound exploration, Lagrangian dual updates, and active/conservative skipping mechanisms.
- Prove gap-independent regret and conservative-skip bounds for routing with combinatorial action spaces and budgeted feedback.

Dynamic and Interpretable Neural Network Models *2024–2025*

- Designed lightweight neural-network models combining data preprocessing, dynamic network tuning, and PDE-inspired regularization.
- Explored interpretable image-classification pipelines that reduce model complexity while preserving competitive performance.

SELECTED PUBLICATIONS

- **Suspicious Alignment of SGD: A Fine-Grained Step Size Condition Analysis**
  Shenyang Deng, **Boyao Liao**, Zhuoli Ouyang, Tianyu Pang, Minhak Song, Yaoqing Yang
  The 37th International Conference on Algorithmic Learning Theory (*ALT*), 2026. **Best Student Paper Award**.
  arXiv: 2601.11789

- **Depth, Not Data: An Analysis of Hessian Spectral Bifurcation**
  Shenyang Deng, **Boyao Liao**, Zhuoli Ouyang, Tianyu Pang, Yaoqing Yang
  IEEE International Symposium on Information Theory (ISIT), 2026.
  arXiv: 2602.00545

SELECTED MANUSCRIPT

- **MAB Budgeted Routing**
  Boyao Liao et al.
  Manuscript in preparation. Studies online shortest-path routing with combinatorial semi-bandit feedback and hard anytime budget constraints; proposes AUSAK with gap-independent regret and conservative-skip guarantees.

SKILLS

Programming	Python, PyTorch, MATLAB, C, Java, LaTeX
Mathematics	Optimization, probability, statistics, linear algebra, real analysis
Languages	Chinese, English

SELECTED COURSES

- Real Analysis, Analytical Geometry, Sequences and Series, Algebra and Combinatorics, Vector Geometry and Linear Algebra, Probability and Statistics, Introduction to Data Analysis, C Programming.

REFERENCES

- **Prof. Yaoqing Yang**
  Dartmouth College
  Research Advisor
- **Prof. Hongyi Jiang**
  City University of Hong Kong
  Research Mentor